WorldConsistMem: Cross-Query Consistency Evaluation for Long-Term Memory in Evolving Worlds

Anonymous authors
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Abstract

Standard long-term memory evaluations score answers one query at a time. A system can therefore look accurate while its answers fail as one coherent world history, so aggregate Acc can overstate joint reconstruction and mis-rank systems.

WorldConsistMem evaluates that discrepancy. Dependent query bundles are grounded in one gold evolving history; machine-checkable constraints $\mathcal{C}$ induce a joint predicate Φ on the predicted answer set. Acc scores local gold match; Bundle Consistency Rate (BCR) is the fraction of bundles with $\Phi = 1$; $\text{Gap}_{\text{query}} = \text{Acc} - \text{BCR}$ measures the gap. We treat BCR as a different evaluation object from Acc, while explicitly accounting for conjunctive strictness via an Acc^n independence diagnostic.

On frozen Option A (102/1,088/7,140), Acc and BCR diverge—including identical Acc with different BCR (0.4905 with BCR 0.219 vs. 0.000) and BM25 Acc = 0.929 with BCR = 0.591 (below Acc^n). On a frozen stratified transfer subset (149 queries / 22 complete bundles), a proprietary-reader check with GPT-4.1 mini yields 45.6% mean query Acc while BCR = 0/22, so substantial item Acc coexists with complete observed bundle-consistency failure under the same H0 protocol. Scope: controlled diagnostic plus bounded transfer checks—not open-world or frontier-model coverage.

1 Introduction

Long-term memory systems must answer what is true now, what was true earlier, how transitions occurred, and which evidence supports those claims. Standard evaluations score these questions one at a time (Wu et al., 2025; 2026; Hu et al., 2025; Tao et al., 2026; Xie et al., 2026; Wang et al., 2026; Hu et al., 2026; Liu et al., 2026). Acc can therefore look high while dependent answers fail as one reconstructed world history (Figure 1).

Central claim. Under known gold evolving histories, item-level Acc systematically overstates whether dependent answers jointly satisfy world-history constraints. We operationalize the missing object with Φ / BCR and $\text{Gap}_{\text{query}} = \text{Acc} - \text{BCR}$, and diagnose conjunctive strictness with an Acc^n baseline (not a generative model).

Approach. WorldConsistMem supplies dependent bundles over one gold history, compiles machine-checkable constraints $\mathcal{C}$, and scores $\Phi = \mathbf{1}[\forall c \in \mathcal{C} : c(\hat{Y})]$ on the predicted answer set. Acc is local gold match; BCR is the mean of Φ. Φ is not "exact-match conjunction" alone: constraints can fail on incompatible transitions/provenance even when many items are locally correct, and can hold for coherent-but-wrong worlds.

Primary + confirmatory evidence. Frozen Option A (102/1,088/7,140) is the primary Acc–BCR divergence case. Separately labelled confirmatory probes include parser re-scoring of frozen Qwen H4, a 36-scenario NL control set, and a proprietary-reader transfer check on a frozen stratified H0 subset (149 queries / 22 complete bundles) with GPT-4.1 mini (Section 7.5).

Contributions.

1. A formal evaluation object for dependency-linked belief consistency (Acc/Φ/BCR/Gap$_{\text{query}}$) with an explicit conjunction diagnostic.
2. WorldConsistMem: controlled evolving-world benchmark with machine-checkable history-derived constraints.
3. Empirical Acc–Φ divergence on the frozen symbolic study, including identical Acc with different BCR.
4. Confirmatory probes: parser re-scoring, bounded NL controls, and an independent proprietary-reader transfer check on complete bundles with conservative handling of truncated outputs—not open-world or frontier-model coverage.

2 Related Work

LTM Acc suites grade per-query competence under long histories (Wu et al., 2025; 2026; Hu et al., 2025; 2026; Wang et al., 2026). State tracking / evolving knowledge recovers values at times or checkpoints (Xie et al., 2026; Liu et al., 2026). Conflict / temporal work shows correctness can diverge from retrieval fitness (Tao et al., 2026). Logical cross-query consistency already separates Acc from global coherence (Salla et al., 2026; Chaudhry, 2026); we do not claim that separation is new in general. Our residual is the evaluation object: persistent memory over evolving multi-entity gold worlds with history-derived validity/transition/provenance/relation constraints scored jointly. Appendix C (Table 8) details closest benchmarks.

3 Cross-Query Consistency in Evolving Worlds

Let $W_0, \ldots, W_T$ be structured world states with events $E_{1:T}$ and $W_t = \text{Apply}(W_{t-1}, E_t)$. The gold history is retained for scoring only. The system receives observations $O_{1:T}$ and holds memory M_T. A dependent bundle $Q_b = \{q_1, \ldots, q_{n_b}\}$ asks interdependent current/historical/transition/relational/provenance/conflict questions about one focal lifecycle. Predictions $\hat{Y}_b = \{\hat{y}_i\}$ are produced by a reader on M_T. Constraints $\mathcal{C}_b$ are compiled from the gold history and query slots.

Operational definitions.

$$\text{Acc}_i = \mathbf{1}[\hat{y}_i \equiv y_i^*] \quad \text{(structured exact match to gold)}, \tag{1}$$

$$\Phi(\hat{Y}_b; \mathcal{C}_b) = \mathbf{1}\left[\forall c \in \mathcal{C}_b :\ c(\hat{Y}_b)\right] \quad \text{(joint world-history consistency)}, \tag{2}$$

$$\text{Acc} = \tfrac{1}{N} \sum_i \text{Acc}_i, \qquad \text{BCR} = \tfrac{1}{|B|} \sum_{b \in B} \Phi(\hat{Y}_b; \mathcal{C}_b), \qquad \text{Gap}_{\text{query}} = \text{Acc} - \text{BCR}. \tag{3}$$

Here N is the number of scored queries and B the set of bundles. Acc$\in [0,1]$ and BCR$\in [0,1]$. If $\mathcal{C}_b = \emptyset$, we define $\Phi = 1$ (Section 5). Acc aggregates query-local gold matches. Φ is a property of the entire answer set under world-history constraints—not mean Acc and not pairwise agreement alone (Figure 1).

Frozen vignette (B2 latest-only). Table 1 shows one CEO-succession bundle from the frozen Option A predictions. Six of eight items match gold (Acc = 0.75), including current CEO and the succession transition, but the system abstains on the historical CEO (and on temporal order). Under $\Phi = \mathbf{1}[\forall c \in \mathcal{C}_b : c(\hat{Y}_b)]$, those abstentions leave required slots unfilled, so transition/contradiction/temporal constraints fail and $\Phi = 0$ for this belief set—hence a BCR contribution of 0. Acc alone does not surface that joint failure. This is a metric-level illustration, not a claim that abstention always zeros BCR, nor a causal diagnosis of memory loss.

Proposition (identical Acc, different BCR). There exist prediction assignments with equal Acc but unequal BCR. Minimal construction. Two bundles of three gold-linked queries

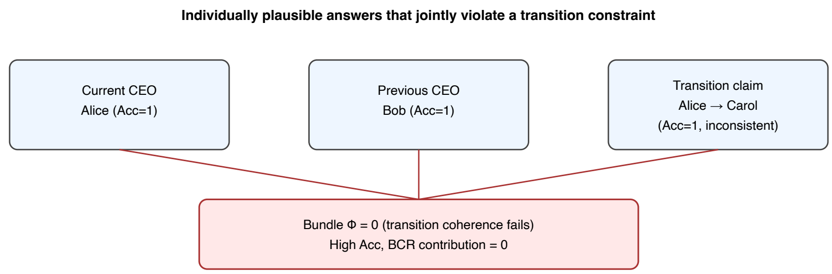

Figure 1: Minimal failure: Acc_i=1 for current/previous CEO, yet Φ=0 on the transition constraint.

Table 1: Frozen. B2 latest-only on one CEO bundle (8 queries; Acc = 0.75, Φ=0). Entities abbreviated; full IDs in the reproducibility extract.

Query role	Gold	Prediction	Correct?	Belief-set effect
Current CEO	P_1	P_1	Y	local Acc hit
Historical CEO	P_0	abstain	N	breaks trans./contrad.
Succession	$P_0 \to P_1$	$P_0 \to P_1$	Y	insufficient alone under $\forall c$
Temporal order	true	abstain	N	fails temporal c

each; $N = 6$. System A: one fully gold bundle and one partial (Acc = 4/6, BCR = 1/2). System B: two Acc hits per bundle but failed transitions (Acc = 4/6, BCR = 0). Empirically, frozen B2 and B3-k3 share Acc = 0.4905 with BCR 0.219 vs. 0.000 (Section 7).

What $\mathrm{Gap}_{\mathrm{query}}$ means—and does not. A large positive $\mathrm{Gap}_{\mathrm{query}}$ means local gold matches occur often while fully coherent bundles are rarer. $\mathrm{Gap}_{\mathrm{query}}$ does not measure retrieval quality alone, factual correctness of a wrong-but-coherent world (consistent_wrong_world), overall memory usefulness, or real-world reliability.

Conjunction penalty (acknowledged). Because Φ requires $\forall c$, BCR is mathematically stricter as $|\mathcal{C}_b|$ and n_b grow. We therefore report a diagnostic baseline $\widehat{\mathrm{AC}}_b = \mathrm{Acc}_b^{n_b}$ (i.i.d. item-correctness approximation) and compare BCR to both Acc and mean $\widehat{\mathrm{AC}}$. A gap Acc − BCR that remains after comparing BCR to $\widehat{\mathrm{AC}}$ cannot be attributed solely to "Acc to the power n." This baseline is not the true error process and does not replace primary BCR.

4 WorldConsistMem Benchmark

WorldConsistMem instantiates Section 3: typed entities, lifecycle events with validity/provenance, observation streams without gold W_t, and dependent bundles with history-derived constraints (Table 2).

Why synthetic? Causal Acc-vs-Φ scoring needs a known gold history and auto-derived constraints. Open dialogue lacks that joint gold without a second noisy annotator. We isolate structural claims (Acc≠BCR) under controlled seed/tier/k/corruptions; ecological validity is not claimed. A confirmatory naturalistic set (Section 7) tests bounded NL surface transfer only.

Table 2: Dataset statistics (scaled freeze, seed 100).

Quantity	Value
Worlds (S/M/L)	102 (34/34/34)
Bundles / queries	1,088 / 7,140
H4 LLM subset	18 / 192 / 1,260

Table 3: Frozen. Corruption: Acc and BCR move independently.

Corruption	Acc	BCR	$\text{Gap}_{\text{query}}$
high_acc_inconsistent	0.849	0.000	0.849
consistent_wrong_world	0.394	1.000	−0.606
transition_inconsistency	0.939	0.600	0.339

5 Evaluation Metrics

We report Acc, BCR, $\text{Gap}_{\text{query}}$, and CSR jointly (Eq. 3). For bundle b, CSR_b is the fraction of applicable constraints in $\mathcal{C}_b$ that the predicted answer set satisfies (with CSR_b=1 if $\mathcal{C}_b$=$\emptyset$). Equivalently, BCR is the fraction of bundles with CSR_b=1 (equivalently Φ=1). CSR does not replace BCR: it resolves floors when no bundle is fully coherent. We also report ConsAcc / PConsAcc grids, family violation rates, abstention/malformed rates, and world-level bootstrap 95% CIs (worlds as primary units). Metrics are computed from frozen predictions by the released evaluators; they do not depend on memory internals.

6 Evaluated Systems and Setup

Systems are reference implementations, not claimed winners: B1 compact flat, B2 latest-only, B3 recency-k, B4 BM25-k, H0 hierarchical (cap150, ret8).

Readers. Symbolic exact-match on the full freeze. LLM Option A: Qwen2.5-3B-Instruct-4bit via MLX; temp 0; max tokens 48; frozen JSON prompt; evidence packs $\leq$3; H4 only. Confirmatory (not Option A): parser re-scoring of frozen Qwen raw outputs; proprietary-reader transfer check with requested alias gpt-4.1-mini on a frozen stratified H0 subset (Section 7.5). Fairness caveats: Appendix D. Frozen vs confirmatory never pooled.

7 Results

Frozen Option A is primary. Confirmatory outputs are labelled and never pooled into Option A tables. Independent units: world for frozen BCR CIs; scenario for transfer CIs.

7.1 RQ1: How large is Acc vs. complete belief consistency?

Corruption. Gold Acc=BCR=1. Controlled corruptions move Acc and BCR independently (Table 3).

Scaled symbolic. Table 4 and Figure 2: B2/B3-k3 share Acc = 0.4905 with BCR 0.219 vs. 0.000; B4 Acc = 0.929 with BCR = 0.591 ($\text{Gap}_{\text{query}}$=0.338). Table 1 shows a concrete B2 belief set with Acc = 0.75 and Φ=0.

7.2 RQ2: Persistence across systems/readers?

Symbolic gaps persist across systems/tiers/ablations (Figure 4; Table 11). Under frozen Qwen2.5-3B H4, all four systems have BCR = 0 while Acc$\in [0.093, 0.185]$ (Table 5); gold H4 yields Acc=BCR=1. Because item Acc is low, the LLM BCR floor is primarily a difficulty/abstention probe, not high-Acc/low-BCR divergence. Confirmatory: re-parsing

Table 4: Frozen. Symbolic scaled (102/1,088/7,140). Brackets: world-level 95% CIs.

System	Acc	BCR	$\mathrm{Gap}_{\mathrm{query}}$	mean CSR
B2 latest-only	0.490	0.219	0.272	—
B3 recency-k3	0.490	0.000	0.490	—
B3 recency-k8	0.567	$0.063_{[0.049,0.084]}$	0.504	$0.507_{[0.483,0.547]}$
B4 BM25-k8	0.929	$0.591_{[0.586,0.596]}$	0.338	$0.881_{[0.870,0.890]}$
H0 LTKM (cap150, ret8)	0.976	$0.844_{[0.820,0.898]}$	0.132	$0.973_{[0.970,0.983]}$

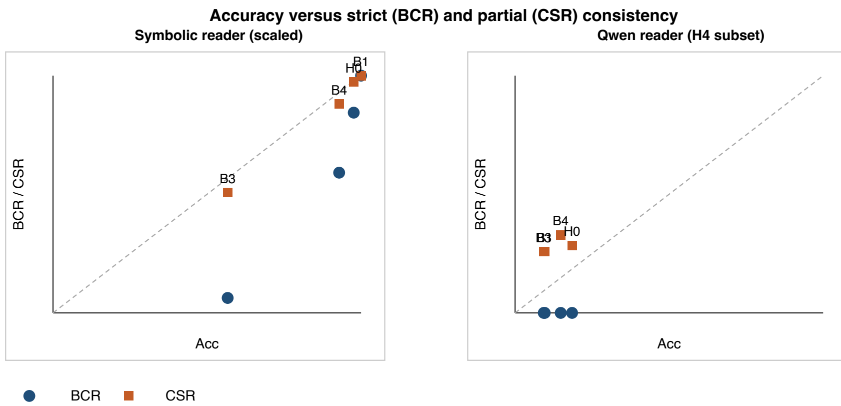

Figure 2: Frozen. Acc vs BCR/CSR.

frozen Qwen raw_output with three extractors leaves Acc/BCR unchanged at BCR = 0 (H0 Acc $\approx$ 0.187; Appendix F). Phi-3.5 confirmatory H4 was blocked (no MLX runtime/weights in this environment).

7.3 RQ3: Only belief-set size, length, abstention, or parsing?

Conjunction diagnostic (confirmatory). On frozen predictions re-scored read-only, B4 has Acc = 0.925, mean Acc^n=0.714, BCR = 0.591: Acc − BCR=0.334 decomposes into Acc−Acc^n (conjunctive) plus Acc^n−BCR = 0.123 beyond the i.i.d. baseline (world-level BCR CI $[0.586, 0.596]$). H0 similarly sits below Acc^n (BCR = 0.844 vs. 0.897). Thus BCR is stricter than Acc, and not identical to Acc^n.

Abstention. Bundles without abstentions often achieve BCR = 1 under symbolic B2/H0; abstaining bundles drive many Φ=0 outcomes. This explains part of the gap without erasing it: Acc still counts remaining correct items (Table 1).

Retrieval / shortcuts (NEW). Family-majority Acc = 0.352 with BCR = 0; B2/B3-k3 Acc-tie under different k (Table 12).

Parsers. Qwen BCR = 0 is stable across strict/relaxed/field parsers, but Acc remains low under all three (Appendix F); parser agreement does not by itself prove semantic memory failure.

7.4 RQ4: Controlled naturalistic transfer (confirmatory probe)

We froze 36 independent NL scenarios (4 queries each; 144 scored queries; seed 20260908) as a separately frozen confirmatory study. Bounded conclusions only: gold Acc=BCR=1; last-event Acc = 0.208 with BCR = 0 (low Acc; difficulty/control probe). Details in Appendix G.

Table 5: Frozen Option A. Qwen2.5-3B-Instruct-4bit; H4. All BCR = 0.

System	Acc	BCR	Abs	Mal
B1 compact	0.096	0.000	0.728	0.007
B4 BM25-k8	0.148	0.000	0.579	0.001
H0 LTKM	0.185	0.000	0.483	0.029

Table 6: Proprietary-reader transfer check on a frozen stratified H0 subset (NEW; not Option A). Acc CIs: bundle-level bootstrap (10,000 resamples). BCR CIs: exact Clopper–Pearson for observed 0/22 (two-sided hi 15.4%; one-sided 95% upper 12.7%). Six GPT-4.1 mini responses hit the 48-token ceiling with truncated JSON and were conservatively counted incorrect (no selective retry).

Reader	Q	B	Acc	Acc 95% CI	BCR	BCR 95% CI	Gap	Abs / Trunc
GPT-4.1 mini	149	22	45.6%	$[40.4, 50.7]$	0/22	$[0, 15.4]$	45.6	41 / 6
Qwen 3B (matched)	149	22	20.1%	$[13.2, 27.1]$	0/22	$[0, 15.4]$	20.1	62 / 7

7.5 RQ5: Proprietary-reader transfer check?

On a frozen stratified transfer subset of the H4/H0 protocol (149 queries in 22 complete bundles; seed 20260909; not Option A), we re-read identical evidence packs with requested alias gpt-4.1-mini (temperature 0; 48-token ceiling; prompt prompt_v3_k3_mt48). An independent smoke test resolved the alias to gpt-4.1-mini-2025-04-14; stored rows record the alias, not resp[model]. Six outputs hit the ceiling with truncated JSON and are counted as incorrect (no selective retry). Table 6 reports the check alongside an exact matched-subset extraction of frozen Qwen H0 predictions for the same 149 IDs.

Substantial query-level Acc can therefore coexist with complete observed bundle-consistency failure, and the Acc–BCR gap transfers from the open-weight H0 evaluation to an independently evaluated proprietary reader under the same system stack (model + frozen prompt + H0/LTKM retrieval + parser + strict Φ). This does not establish failure across frontier models.

8 Consistency Failure Analysis

Table 1 is the frozen symbolic vignette. Under Qwen H4, transition constraints fail on 192/192 bundles (Table 7)—the same joint-failure class as Figure 1. Family rates: Figure 3; Table 9.

9 Discussion and Limitations

9.1 Alternative explanations

- Retrieval only: Acc-tied B2/B3-k3; shortcuts Acc>0 with BCR=0.
- Conjunction only: BCR lies below Acc^n for B4/H0 (residual beyond i.i.d.).
- Abstention only: abstention predicts many Φ=0 cases but Acc still overstates joint success on partial hits.
- Parser only: Qwen BCR=0 stable across three parsers at low Acc; the proprietary-reader check reaches Acc = 45.6% with BCR = 0/22 under the same ceiling/parser protocol, so the gap is not explained by a Qwen-only floor.
- Length/tier only: frozen gaps persist across tiers; NL transfer uses short narratives as a control probe.
- Consistency$\neq$factual correctness: consistent_wrong_world; narrative heuristic Acc<1 with BCR=1.

Table 7: Frozen. Qwen violation multiplicity (H4).

System	Mean viol./bundle	Frac. one	Frac. multiple
B4 BM25	3.33	0.177	0.823
H0	3.50	0.000	1.000

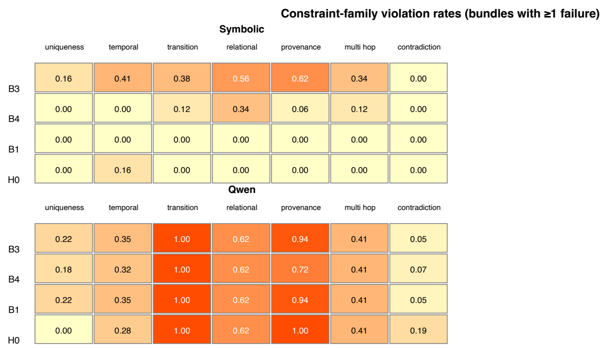

Figure 3: Frozen. Constraint-family violation rates.

9.2 Threats to validity

Synthetic ontology limits open-world claims; naturalistic NL transfer remains a heuristic control probe. Reader coverage: frozen Qwen2.5-3B H4 plus one proprietary-reader transfer check (GPT-4.1 mini on 22 bundles). GPT-4.1 mini is proprietary but non-frontier; only one such reader was evaluated; six outputs were truncated at the predetermined ceiling. This is a system-level check (retrieval + prompt + model + parser + scorer), not model-only causation. Seed diversity for B3 mini-sweeps is limited. Metric scope: BCR is history-constraint compatibility, not usefulness/safety. External human validation is absent.

9.3 BCR vs CSR

When BCR floors, report CSR/violation multiplicity (Table 16); do not replace BCR with a favorable partial cell.

10 Conclusion

Demonstrated. Acc and BCR measure different objects. In frozen Option A, Acc overstates joint world-history consistency and can fail to separate Acc-tied systems; an Acc^n diagnostic shows BCR is not solely "Acc to the power n." On a frozen stratified transfer subset, GPT-4.1 mini achieves 45.6% mean query Acc with BCR = 0/22 under the H0 system stack, so the Acc–consistency gap transfers to an independently evaluated proprietary reader.

Significance (bounded). Acc-only evaluation of long-horizon memory can mis-rank systems and hide joint failures that matter for historical reconstruction—in this controlled setting.

Scope. We do not claim ecological validity, frontier-model coverage, or deployed-agent generalization. Consistency is not factual correctness.

Next validation. Expand proprietary-reader coverage and meaningful-Acc NL readers under the same complete-bundle protocol.

AI Use Statement

This statement is required by ICLR 2027 and does not count toward the page limit. We used generative AI coding assistants for manuscript editing, LaTeX packaging, experiment scripting assistance, analysis code, and repository tooling. We did not use generative AI to invent experimental results, numerical claims, citations, or benchmark labels. All headline numbers were checked against frozen CSVs or labelled confirmatory re-evaluations. We reviewed AI-assisted text for accuracy and take responsibility for the final content.

Ethics Statement

WorldConsistMem uses fully synthetic worlds and machine-checkable constraints; the naturalistic transfer set uses fictional names only. No human subjects or personal data are involved. Real-world accountability claims are not made from this evidence.

Reproducibility Statement

We release seed-deterministic generation (seed 100; 102/1,088/7,140), frozen symbolic and Qwen predictions, metric definitions, decoding freeze, confirmatory scripts under results/confirmatory/, and reference implementations. Labelled confirmatory/NEW analyses are stored separately from Option A. Identifying URLs are omitted for double-blind review. Appendix A details protocols and checksums.

Acknowledgements

Omitted for double-blind review.

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A Additional Reproducibility Details

Dataset. Scaled WorldConsistMem is generated with seed 100. Smoke validation confirms gold Acc = BCR = ConsAcc = 1 and Acc–Φ separation under controlled corruptions.

Readers. Symbolic exact-match scoring is used for the full scaled study. The LLM condition uses Qwen2.5-3B-Instruct-4bit with frozen short-form JSON decoding (temperature 0; max tokens 48) on the H4 subset (18 worlds / 192 bundles / 1,260 queries).

Systems. Headline reference systems: B1 capacity-limited compact flat; B3 recency retrieval; B4 BM25 retrieval; H0 structured hierarchical reference. Ablations and capacity sweeps are reported in the released artefacts.

Metrics. Primary consistency rate: BCR. Complementary: CSR, ConsAcc / PConsAcc grid, $\mathrm{Gap}_{\mathrm{query}}$, family violation rates. World-level bootstrap 95% CIs accompany BCR/CSR (and world-mean Acc) in frozen CSVs.

Labelled rescue sanity (not Option A). Re-evaluating gold answers on the H4 subset yields Acc=BCR=ConsAcc=1. NEW robustness (seeds; density; distractors; shortcuts; retrieval-budget) is in Section 7.3. Confirmatory artefacts live under results/confirmatory/ and results/new/stronger_reader_pilot/ (proprietary-reader transfer; Appendix H). The Table 1 extract is results/new/b2_vignette_ss0100_ceo0.json.

Reader coverage (updated). Option A completed Qwen2.5-3B-Instruct-4bit. A separately labelled proprietary-reader transfer check used requested alias gpt-4.1-mini on a frozen stratified H0 subset (149/22). Phi-3.5 local MLX runs remain blocked in some environments. We do not claim frontier-model coverage.

Checksum anchors. Scaled manifest sha256 37aa8604c3e6d87691332f66d2f7c0ab93578f2fe468bc29f7e8217c1675af3f; symbolic system_results.csv sha256 eb5bd5fa87496586bd830e5f685b9950e119f9fd0ac491194b3f003343f9a2a8; smoke dataset canonical sha256 3aa67ed1b6a39be3d9e1805d14f7d5ce7cc05f4655bdce0bc4ff49900c14c7ab.

B Artifact Checklist

Released artefacts accompanying this submission include: dataset generators and manifests; symbolic system results and predictions; Qwen H4 system/query-family/constraint-family results; partial-consistency recomputes; corruption-suite metrics; labelled NEW robustness JSON; and smoke/reproduction scripts. Manuscript figures are static PDFs; figure-drawing source scripts were not present in the experiment tree. Anonymized supplementary material is provided with the OpenReview submission; identifying repository URLs are omitted from this PDF.

C Extended Related Work

C.1 Long-term memory evaluation

LongMemEval evaluates chat assistants on long-term interactive memory through five ability axes—information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention—with answer accuracy as the primary metric under long-history and long-context protocols (Wu et al., 2025). LongMemEval-V2 extends the agenda toward "experienced colleague" competence, adding dynamic state tracking, workflows, and premise awareness, again with per-question accuracy (and latency-oriented reporting) as the headline outcomes (Wu et al., 2026). MemoryAgentBench evaluates agent memory via incremental multi-turn interactions across competencies such as accurate retrieval, test-time learning, long-range understanding, and selective forgetting / FactConsolidation after edits, scored primarily by task or answer accuracy (Hu et al., 2025). EverMemBench targets long-horizon collaborative, multi-party dialogues and reports fine-grained recall, memory awareness, and profile understanding under QA accuracy protocols (Hu et al., 2026). EvoMemBench organizes agent memory from a self-evolving perspective with a scope×content taxonomy and reports task success or accuracy (and efficiency) across methods compared with long-context baselines (Wang et al., 2026). Taken together, this line provides mature protocols for long-horizon Acc and ability factorization; the measurement gap we target is joint coherence of dependent answers, not another Acc axis.

These benchmarks primarily evaluate individual query or task outcomes. WorldConsistMem does not replace ability Acc suites. It additionally evaluates whether answers within a dependent bundle jointly satisfy constraints derived from one world history. The evaluation unit is the interdependent answer set $\hat{Y}_b$ under $\Phi(\hat{Y}_b; \mathcal{C}_b)$, not a single graded QA item.

C.2 Dynamic and evolving knowledge

Evolving worlds are not novel by themselves. DynamicMem constructs long-horizon personal memory in real-world settings and evaluates temporal checkpoint reconstruction and related state-tracking accuracy for evolving user attributes, habits, and preferences (Xie et al., 2026). WorldMemArena evaluates multimodal agent memory through action–world interaction, diagnosing write, maintain, retrieve, and use stages under lifelong evolution and agentic execution protocols (Liu et al., 2026).

It is important to distinguish three notions that are easy to conflate. Tracking changing state asks whether the system recovers the correct value at a time or checkpoint. Profile or lifecycle consistency asks whether personal attributes, stage metrics, or write/use pipelines remain coherent for a user or agent trajectory. Cross-query world-history consistency asks whether interdependent answers about current state, historical state, transitions, relations, and provenance jointly satisfy constraints induced by one shared multi-entity gold history.

C.3 Conflict, temporal validity, and provenance

MemConflict evaluates long-term memory systems under dynamic, static, and conditional memory conflicts, combining black-box answer correctness with white-box retrieval and ranking diagnostics, and showing that answer correctness can diverge from retrieval fitness (Tao et al., 2026). These lines diagnose conflict handling, version validity, and temporal updates at the query, retrieval, or competency level. The WorldConsistMem contrast is evaluation grain and joint structure rather than denial of those pathologies: a conflict- or update-correct single answer can still participate in an inconsistent bundle.

C.4 Cross-query logical consistency

Salla et al. (2026) study case-file logical consistency across interdependent queries and introduce set-level metrics including Case Satisfiability and SetConsRate. Chaudhry (2026) maintain persistent symbolic belief states verified with SMT-style checking and release LogicBench-Cross. Machine-checkable constraint evaluation over multi-query answer sets

Table 8: Closest related benchmarks and adjacent consistency work. Cells use Y / P / N for whether the feature is a primary evaluation object as operationalized in WorldConsistMem.

Work	LTM	Evol.	Bundles	Gold hist.	Trans.	Prov.	CQ constr.	Strict BCR	Partial CSR	Primary metric
LongMemEval	Y	P	N	P	P	N	N	N	N	Ability Acc
LongMemEval-V2	Y	Y	N	P	P	N	N	N	N	Acc (+ latency)
MemoryAgentBench	Y	P	N	P	P	N	N	N	N	Competency Acc
EverMemBench	Y	Y	N	P	P	N	N	N	N	Recall / profile Acc
EvoMemBench	Y	P	N	P	P	N	N	N	N	Task Acc
MemConflict	Y	P	N	P	P	P	P	N	P	Conflict Acc + retrieval
DynamicMem	Y	Y	P	Y	P	N	P	N	P	Checkpoint Acc
WorldMemArena	Y	Y	P	P	P	P	N	N	P	Lifecycle Acc
SetCons metrics	N	N	Y	P	P	N	Y	Y	P	SetCons / satisfiability
LogicVault	N	N	Y	P	P	N	Y	Y	P	Logical / SMT
WorldConsistMem	Y	Y	Y	Y	Y	Y	Y	Y	Y	Acc+BCR+CSR

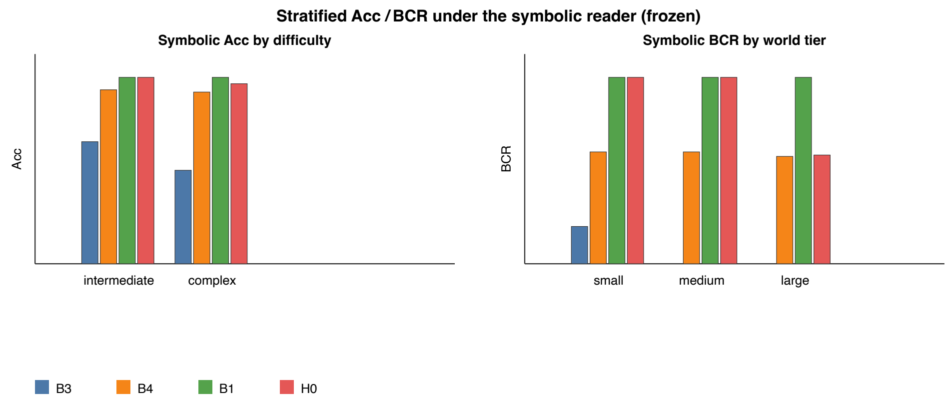

Figure 4: Symbolic Acc by difficulty and BCR by world tier.

is therefore not universally new. WorldConsistMem's residual is application domain and constraint origin: persistent memory systems over evolving multi-entity worlds with world-history-derived constraints.

D Fairness Caveats

Comparisons are informative but not perfectly matched:

- Internal structure differs (flat list vs multi-store H0).
- Capacity accounting charges H0 metadata (graph, provenance, indices) against the same logical-record budgets used for flat stores; absolute budgets therefore tax structured metadata.
- Retrieval budgets (k vs store-routed packs) are matched numerically where stated, but lexical BM25 is not identical to H0's routing.
- Deterministic vs LLM readers are separate conditions; symbolic Acc/BCR must not be pooled with Qwen Acc/BCR.
- Compact flat can retain focal lifecycle events by dropping padding, which can yield very high symbolic Acc/BCR at medium budgets; this is a fairness caveat for matched-budget interpretation, not a general ranking of store designs.

E Additional Figures and Tables

F Qwen parser sensitivity (confirmatory)

Re-scoring frozen H4 raw_output (no re-inference). Model: mlx-community/Qwen2.5-3B-Instruct-4bit. Across strict JSON, relaxed JSON, and answer-field regex, H0 Acc ≈ 0.187

Table 9: Constraint-family violation rates (headline systems).

	Symbolic				Qwen		
Family	B3	B4	B1	H0	B1/B3	B4	H0
uniqueness	0.156	0.000	0.000	0.000	0.219	0.177	0.000
temporal	0.406	0.000	0.000	0.156	0.354	0.323	0.281
transition	0.375	0.120	0.000	0.000	1.000	1.000	1.000
relational	0.562	0.343	0.000	0.000	0.625	0.625	0.625
provenance	0.625	0.057	0.000	0.000	0.938	0.724	1.000
multi_hop	0.344	0.124	0.000	0.000	0.406	0.406	0.406
contradiction	0.000	0.000	0.000	0.000	0.052	0.073	0.188

Table 10: Query-family Acc under Qwen (H4; frozen; three-decimal display).

Family	B1	B3	B4	H0
current_state	0.010	0.010	0.083	0.542
historical_state	0.162	0.162	0.089	0.547
transition	0.000	0.000	0.000	0.000
multi_hop	0.000	0.000	0.052	0.000
provenance	0.188	0.167	0.417	0.000
temporal_order	0.128	0.128	0.205	0.308
contradiction	0.083	0.083	0.167	0.000
counterfactual	0.000	0.000	0.000	0.000
conflict_validity	0.462	0.462	0.462	0.000

and BCR $= 0$; B4 Acc ≈ 0.147 and BCR $= 0$; B3 Acc ≈ 0.075 and BCR $= 0$. Abstention rates $\approx$0.48–0.70; malformed rates $\leq$0.029. Cache-wide error channels: abstention $\approx$54%, substantive/unchecked $\approx$44%, malformed $\approx$1.5%. A fixed raw-output sample (seed 20260908; n=12 H0 rows) shows majority three-parser agreement; occasional field-regex recoveries of malformed strings do not change BCR. Low Acc means this rejects a single-parser artefact without supporting high-Acc Acc–BCR divergence for Qwen.

G Naturalistic transfer details (confirmatory)

Independent scenarios n=36; queries/scenario = 4; total queries = 144; scenarios sha256 b84dfa83…. Definitions: gold oracle emits gold answers; last-event uses only the last narrative sentence (current-like entity; abstains on historical/transition/temporal); entity-prior collapses to a P1/LOC1 shortcut; narrative-heuristic is full-narrative regex extraction. Scenario-level Acc CIs: last-event $[0.174, 0.236]$; gold $[1, 1]$. Role Acc (last-event): current 0.833, others 0. Correct-query counts (last-event): 0 in 6 scenarios, 1 in 30. Conjunction diagnostic (reference only): last-event $p^4 \approx 0.0019$ and role-product = 0, so observed BCR = 0 is consistent with low-p conjunction rather than a high-Acc residual. Narrative heuristic Acc = 0.875 with BCR = 1 (negative $\mathrm{Gap}_{\mathrm{query}}$; consistent-but-not-fully-factual). No LLM reader completed on these scenarios.

H Proprietary-reader transfer check (NEW)

Sampling. Frozen stratified transfer subset of H4/H0: deterministic proportional allocation by (tier $\times$ difficulty), seed 20260909, complete bundles only (22 bundles / 149 queries). Manifest sha256 dd9790c8…. Primary report sha256 e3ac10ae….

Protocol. Requested alias gpt-4.1-mini; independent smoke resolution gpt-4.1-mini-2025-04-14; cache rows store the alias. Temperature 0; max tokens 48; prompt prompt_v3_k3_mt48; H0/LTKM evidence; same JSON parse. Statuses: 102 ok / 41 abstain / 6 malformed. All six malformed have completion_tokens=48 (truncated JSON); counted incorrect; no selective retry.

CIs. Acc: bundle-level bootstrap, 10,000 resamples, seed 20260911, percentile interval. BCR = 0/22: exact Clopper–Pearson two-sided $[0, 15.4\%]$; one-sided 95% upper 12.7%. Does not prove population BCR is exactly zero.

Table 11: Frozen. Symbolic H0 component ablations (reprinted from freeze CSV). Large Acc–BCR gaps under archive/temporal/graph removals.

System	Acc	BCR	$\mathrm{Gap}_{\mathrm{query}}$
H0 unconstrained	1.000	1.000	0.000
H0 cap150 ret8	0.976	0.844	0.132
H–archive	0.876	0.375	0.501
H–temporal-validity	0.876	0.375	0.501
H–graph	0.938	0.594	0.344
H–provenance	1.000	1.000	0.000
H–conflict	0.800	0.812	−0.012

Table 12: Labelled NEW. Retrieval-budget sensitivity (seed 100; 6 worlds/tier; 18/192/1,260; symbolic). Identical Acc with different BCR for B2 vs. B3-k3.

System	Acc	BCR	$\mathrm{Gap}_{\mathrm{query}}$
B2 latest-only	0.4905	0.2188	0.2717
B3 recency-k3	0.4905	0.0000	0.4905
B3 recency-k5	0.5095	0.0208	0.4887
B3 recency-k8	0.5667	0.0625	0.5042
B4 BM25-k3	0.7968	0.3750	0.4218
B4 BM25-k8	0.9317	0.5938	0.3379

Matched Qwen. Exact same 149 query IDs from frozen run2 H0 predictions (sha256 5905b081…); Acc = 20.1%, BCR = 0/22. Analysis outputs under results/confirmatory/proprietary_reader_transfer/ (not pooled with Option A).

Family Acc (GPT-4.1 mini). current 17/22; historical 22/22; contradiction 6/6; temporal_order 7/11; counterfactual 6/11; conflict_validity 5/11; multi_hop 5/22; transition 0/22 (incl. 6 truncations); provenance 0/22. BCR is not reported by family (bundle-level metric).

Table 13: Labelled NEW. Shortcut baselines (seed 100; 6 worlds/tier; symbolic scoring). Family-majority uses oracle gold marginals by query family.

Baseline	Acc	BCR	$\text{Gap}_{\text{query}}$
Empty answers	0.000	0.000	0.000
Family-majority (gold marginals)	0.352	0.000	0.352
Family-shuffle of gold answers	0.280	0.000	0.280
Gold oracle	1.000	1.000	0.000

Table 14: Labelled NEW. Seed sweep summary (symbolic B3/B4 k=8; 6 worlds/tier; seeds 100–105). B3 identical across seeds (limited diversity).

System	Acc range	BCR	$\text{Gap}_{\text{query}}$ range
B3 recency-k8	0.567–0.567	0.0625	0.504–0.504
B4 BM25-k8	0.929–0.932	0.5938	0.335–0.338

Table 15: Labelled NEW. Generator controls (symbolic). Density: 8 medium worlds, seed base 200. Distractors: seed 100, wpt=6.

Control	Setting	System	Acc	BCR
Relation density	0.0 / 1.0	B4	0.917 / 0.909	0.600 / 0.600
Relation density	0.0 / 1.0	B3	0.545 / 0.545	0.000 / 0.000
Distractors	on / off	B4	0.932 / 0.932	0.594 / 0.594
Distractors	on / off	B3	0.567 / 0.567	0.062 / 0.062

Table 16: Evaluation coverage and hard limits.

Dimension	Coverage
Paper shape	Benchmark + evaluation methodology
Symbolic reader	Full scaled (102 / 1,088 / 7,140)
LLM reader	Qwen2.5-3B-Instruct-4bit; H4 subset only
Stronger LLM readers	Proprietary GPT-4.1 mini transfer (22 bundles); not frontier coverage
Architecture superiority	Unsupported under Qwen (all BCR = 0)
Ecological validity	Not claimed (synthetic worlds)
Primary consistency metric	BCR (strict); CSR complementary
Labelled robustness	Seeds/density/distractors/shortcuts/retrieval-budget (separate)
B3 seed diversity	Identical Acc/BCR across seeds 100–105 (not replication)
Proprietary transfer	Frozen stratified H0 subset; Acc 45.6%; BCR 0/22